



ROBOTICS FOR TEENS

ROOKIE | MIDTERM REVIEW

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- **Term 1 Review: Intro, System, Computer, Robot, AI**
- **Electronics, Voltage, Current, Circuit**
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DEFINITIONS

- **MACHINE** is a physical device to do mechanical works and convert input energy to a useful work.
 - Ex.: Gear, Lever, Bicycle, ...
- **AUTOMATION** means doing jobs automatically without human interactions, and system makes decision and do tasks by itself.
 - Ex.: Elevator, Production Lines (PLC), Robots, ...
- **ROBOT** is a machine capable of carrying out a complex series of actions automatically, especially one programmable by a computer.
 - Ex.: Mars Rover, Amazon UAVs, Boston Dynamics Atlas, Tesla Autopilot Cars, ...

SYSTEM

- **SYSTEM** is a group of parts or elements working together to perform a specific function or purpose. (Subsystem, Closed/Open Loop)
 - Ex.: Gearbox, GPS, Nervous System, Islamic Revolution, ...
- **COMPONENTS** of a system are: Input, Process, Output & Feedback (Optional).
 - Ex.: GPS, which receive satellite signal and orbital parameters, then use this information and trilateration to calculate a user's exact location.
- **FEEDBACK** is information about output of a system that is used to adjust its behavior, maintain balance, accuracy, or improvement over time. (Positive/Negative)
 - Ex.: AC temp. sensor, Blood glucose level, YouTube comments, Robot Eyes, ...

COMPUTER

- **COMPUTER** is an electric machine (system) that processes data and performs tasks according to instructions. (process, store data, communicate, do arithmetic and logic, ...)
 - Ex.: Desktop, Laptop, Mobile, Server, IoT and Embedded, ...
- **COMPONENTS** CPU, GPU, RAM, SSD, HDD, Motherboard, Network, Power, Monitor, ...
 - CPU: Cache, Core, GHz – RAM: GB, DDR5, MHz – GPU: VRAM, GDDR6, ...
- **EVOLUTION** from vacuum tubes to nano scale semiconductor transistors
 - Remember benchmark tests and shopping tips.

ROBOT

- **SENSOR** collects data from the environment. They are robot's eyes, ears and skin!
(Robotics, Robot vs Human, Robot as a System, ...)
 - Ex.: IR, Camera, Ultrasonic, Microphone, Gas detector, PH sensor, Touch and pressure, ...
- **ACTUATOR** allows the robot to move or interact with the world, like muscles.
 - Ex.: Motors like DC brushed, BLDC, AC, Magnetic relays, solenoids, ...
- **MICROCONTROLLER** is a compact computer in an IC, is robot's brain.
 - Ex.: AVR, ARM, FPGA, ASIC, SoC, Arduino, ESP32, STM32, ...

ARTIFICIAL INTELLIGENCE

- **ARTIFICIAL INTELLIGENCE** is the capability of computational systems to perform tasks typically associated with human intelligence
 - Ex.: YouTube video recommendation, ChatGPT, Uber autonomous, ...
- **MACHINE LEARNING** concerned with the development and study of statistical algorithms that can learn from data.
 - Ex.: Autonomous vehicles, LLMs like Gemini, Google Translate, ...
- **DATA SCIENCE** is a concept to unify statistics, data analysis, informatics, and their related methods to understand and analyze actual phenomena with data.
 - Ex.: Home recommendation, Weather forecast, Disease prediction, ...

ELECTRONICS

- **ELECTRICITY** is the movement of electrons through a material.
 - Ex.: Static/Flow – Electrons/Protons – Conductor/Semiconductor/Insulator
- **VOLTAGE (V^v)** the “push” that moves electrons. Force push or pull between charged particles.
 - Ex.: FORCE push or pull between charged particles in a FIELD. AC/DC.
- **CURRENT (I^A)** flow of electrons.
 - Ex.: Pure electrons or charged ions. AC/DC. Resistivity(R^Ω)/Conductivity.

RESISTOR

- **RESISTANCE** is how much a material opposes electric current. (More Res => Less Current)
 - Extra: Ohm's Law, $K\Omega$, $M\Omega$, Different Material, Colors and Codes, Multimeter
 - Circuit is a closed path where current flows.
- **TYPES** are fixed, variable or special
 - Ex.: Carbon or metal films, Potentiometer, LDR, Thermistor, ...
- **USED** for protecting circuit with limiting the current and dividing the voltage
 - EXTRA: Series/Parallel components, Open Circuit/Short Circuit, SMD Components, Datasheet

CAPACITOR

- **CAPACITOR** stores and releases electrical energy.
 - Ex.: ...
- **TYPES** are electrolyte, fixed, variable, ...
 - Ex.: ...
- **USED** to stabilize the voltage ripple, multiply the voltage, dc removal, ...
 - Ex.: Used in power supplies, telecom circuits, timing circuits, ...

DIODE

- **SEMICONDUCTOR** is a material that can be altered to conduct electrical current or block its passage.
 - Ex.: Silicon, Germanium, Gallium Arsenide, ...
- **DIODE** is a two-terminal electronic component that conducts electric current in one direction.
 - Ex.: LED, Photodiode, 1N4001, ...
- **USED** in rectifiers, demodulation, logic circuits, lighting, ...
 - Ex.: Power Adapters, Radios, Lights, ...

TRANSISTOR

- **TRANSISTOR** is a semiconductor device used to amplify or switch electrical signals and power and one of the basic building blocks of modern electronics.
 - Ex.: NPN BJT like BC547, N-CH MOSFET like IRF630, ...
- **INTEGRATED CIRCUIT** is a compact assembly of electronic circuits formed from various electronic components like transistors, resistors and capacitors.
 - Ex.: 7805 voltage regulator, 7400 NAND gate, Intel Core i-9 Ultra CPU, ...
- **PINOUT** for a sample NPN BJT transistor (like BC547): Collector, Base, Emitter
 - Ex.: You can find transistors in nearly every electronic device.

UNITS

PREFIX	VALUE
M (Mega)	$10^6 = 1\ 000\ 000$
K (Kilo)	$10^3 = 1\ 000$
m (milli)	$10^{-3} = \frac{1}{1000}$
μ (micro)	$10^{-6} = \frac{1}{1\ 000\ 000}$
n (nano)	$10^{-9} = \frac{1}{1\ 000\ 000\ 000}$

COMPONENT	UNIT
Resistor (R)	Ω or R (Ohm)
Capacitor (C)	F (Farad)
Voltage (V)	V (Volts)
Current (I)	A (Ampere)

EXAMPLE	RESULT
1 KΩ	1000 Ω = 1000 R
100 μF	0.000 1 F = 0.1 mF

PROTEUS CAD

- **COMPUTER-AIDED DESIGN** is the use of computers to aid in the creation, modification, analysis, or optimization of a design.
 - Ex.: Altium Designer, Proteus, AutoCAD, SolidWorks, ...
- **PROTEUS** is a leading EDA software including schematic capture, advanced simulation, PCB auto-routing, ...
 - TIP: Electronic Design Automation (EDA) or Electronic Computer-Aided Design (ECAD)
- **ELECTRONICS CIRCUIT SIMULATORS**
 - Ex.: OrCAD (Cadence) PSPICE, TinkerCAD Circuits, LTSPICE, QSPICE, Qucs-S, ...

PROTEUS COMPONENTS

COMPONENT	KEYWORD	LIBRARY	SYMBOL
Resistor	RES	DEVICE	
Capacitor	CAP	DEVICE	
LED	LED-Blue	ACTIVE	
Transistor	BC547	BIPOLAR	
Battery	Battery	ACTIVE	
Diode	1N4007	DIODE	
Buzzer	Buzzer	ACTIVE	

HANDS-ON EXPERIENCE

- I'll add example circuits here later...